DBA-W Sector Framework

Design Basis for Water Sector Infrastructure

Establishing a Sector-Wide Design Envelope for Critical Water Systems

Design Basis and Military Action Extension — Sector Framework Document.
Issued Under: DBA-MA Parent Framework, IAEA L1 v1.7

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

1. Purpose and Scope

This document establishes a Design Basis for water sector infrastructure. Its purpose is to define the envelope of credible adverse events that water sector systems — drinking water treatment and distribution, wastewater collection and treatment, and the operational technology that governs both — must be demonstrated to withstand. That envelope does not currently exist in a concise, sector-wide form. This document creates it.

The water sector is designated Critical Infrastructure under the National Infrastructure Protection Plan (NIPP). Every sector within that designation claims to be critical. Few have translated that claim into a design basis — a formal statement of the events the infrastructure must be designed and shown to withstand. The nuclear sector has one, developed over decades of regulatory practice. The electricity sector has functional equivalents in its reliability standards. The water sector does not. Risk assessments exist. Vulnerability analyses exist. Sector-specific plans exist. A design basis does not.

This document fills that gap at the sector level. It draws on the established risk-based materials — principally AWWA J100-21, the Water Sector Specific Plan, and the 2023 America's Water Infrastructure Act Operational Guide — and reframes their analytical content as a design envelope. The objective is a document that functions as the water sector's foundational safety standard: the set of postulated adverse events against which sector infrastructure must be assessed, and from which protective requirements derive.

This document is the initial instrument in the DBA-W series, which is a member of the DBA-MA family of sector-specific design basis frameworks. The DBA-MA family extends the Design Basis Accident methodology — the foundational analytical framework of nuclear safety — across critical infrastructure sectors. For the water sector, the development sequence is: establish the Design Basis first; extend it to military action and other threat categories second. This document executes the first stage. Extensions — including military action and a fully developed tiered standards framework — are addressed in subsequent DBA-W series documents.

This document is also intended to serve as a professional reference for engineering firms and technical consultancies that design, assess, and advise water sector clients. In the absence of a sector-wide regulator with authority to mandate a design basis, engineering practice is the nearest-term mechanism for propagating design-basis thinking across the sector. Firms that adopt the DBA-W framework as their internal standard for project design, resilience assessment, and client advisory work apply that standard across their entire client portfolio — establishing a de facto design basis whose reach exceeds what regulation currently compels. Section 2.5 addresses this pathway in detail.

The scope of this document is the continental United States water sector. It covers:

- Community water systems serving populations of 3,300 or more, subject to Safe Drinking Water Act requirements and America's Water Infrastructure Act resilience mandates.
- Publicly owned treatment works (POTWs) that manage wastewater collection and treatment.
- The operational technology — supervisory control and data acquisition (SCADA) systems, instrumentation and control networks, remote monitoring infrastructure — that governs both.

Privately owned water utilities and smaller systems serving populations below the AWIA threshold are within the sector's consequence footprint but outside the primary regulatory scope of this document. The design basis established here is intended to be extensible to those systems as the framework matures. Reference event DBA-W-RE-1 (Section 5.3) found that the July 2026 campaign reached utilities across the full size range, including systems below this threshold; a bounded minimum-requirement set for sub-threshold systems is accordingly established as a forthcoming DBA-W instrument rather than left to extensibility alone.

2. The Water Sector and Its Regulatory Context

Understanding what the water sector has — and does not have — in terms of regulatory foundation is necessary before a design basis can be constructed. The sector's regulatory terrain is fragmented, primacy-divided, and consequence-light relative to the magnitude of the infrastructure it governs.

2.1 Federal Regulatory Framework

The Safe Drinking Water Act (SDWA) is the primary federal statute governing public water systems. It establishes maximum contaminant levels and treatment technique requirements and is administered by the Environmental Protection Agency (EPA). The SDWA's focus is on water quality — specifically, contaminant concentrations and treatment performance. It does not establish infrastructure resilience standards in the engineering sense. It does not define an envelope of adverse events that a water system must be designed to withstand.

The America's Water Infrastructure Act of 2018 (AWIA) represents the most significant expansion of water sector security requirements in recent history. It requires community water systems serving populations above 3,300 to conduct risk and resilience assessments and develop emergency response plans. The assessments must address malevolent acts and natural hazards. However, AWIA does not prescribe what the risk and resilience assessment must conclude, what protective measures must result from it, or what performance standard the system must demonstrate. It is a process requirement, not a performance standard. AWIA created the obligation to assess; it did not create a design basis.

The Clean Water Act (CWA) governs wastewater discharge and treatment through the National Pollutant Discharge Elimination System (NPDES) permit program. Like the SDWA, the CWA's primary focus is on environmental quality, not infrastructure resilience. Wastewater treatment works are subject to fewer resilience requirements than drinking water systems, despite their equivalent consequence potential in failure scenarios.

2.2 State Primacy and Fragmentation

Primary enforcement authority for the Safe Drinking Water Act is delegated to states through primacy agreements. Every state except Wyoming and Washington, D.C., holds primacy for public water system oversight. This means that the operational regulatory relationship for most water systems is with a state agency, not the EPA directly. The consequence of primacy delegation is structural fragmentation: fifty different regulatory regimes, with varying levels of technical capacity, enforcement resources, and resilience-focused requirements layered atop the federal minimum.

The absence of a sector-wide design basis is partly a product of this fragmentation. No single regulatory authority is positioned to mandate a national design envelope. The EPA can set minimum standards; it cannot compel states to adopt a design basis framework that exceeds those minimums. This creates the gap that this document addresses: a national-level design basis to inform state implementation, federal guidance, and, ultimately, regulatory or legislative action.

2.3 Sector Specific Plan and NIPP Framework

The Water and Wastewater Systems Sector Specific Plan (SSP), maintained jointly by EPA and the sector's Government Coordinating Council and Sector Coordinating Council, establishes the sector's framework for critical infrastructure protection under NIPP. The SSP identifies risk management goals, partnership structures, and coordination mechanisms. It provides a useful catalog of threat categories and a framework for sector-level coordination.

The SSP operates in risk-management space, not design-basis space. It identifies threats and establishes processes for managing them. It does not define the set of adverse events against which water infrastructure must be demonstrated to be resilient. The SSP is a necessary input to the design basis developed in this document, but it is not a substitute for it.

2.4 The Regulatory Gap

The nuclear sector has a design basis because its regulatory history compelled one. The Atomic Energy Act, the Energy Reorganization Act, and decades of NRC rulemaking produced a body of requirements that define, with engineering specificity, the events a nuclear facility must be designed to withstand and the safety systems that must function during those events. That body of requirements is the product of a regulatory system that treats the severity of consequences as the determinant of regulatory rigor.

The water sector has not had an equivalent driver. Drinking water contamination events — even major ones — have not produced the kind of catastrophic, visible, immediate consequence that would force a design basis onto the regulatory agenda. Cyber intrusions into water sector operational technology have occurred with increasing frequency and growing sophistication, but no event has yet resulted in mass casualties. The regulatory system responds to consequences, not to the proximity of potential consequences.

This document is written against that backdrop. It does not wait for a catastrophic water sector event to compel a design basis. It constructs one from the risk and resilience analytical work the sector has already done, on the premise that the consequences of water sector failure — public health at the population scale, cascading effects on every other CIKR sector — are severe enough to warrant design-basis thinking before an event forces it.

2.5 Engineering Practice as a Pathway to Sector-Wide Standards

The regulatory gap described in Section 2.4 defines the problem; it does not define the only pathway to a solution. Formal regulation is one mechanism by which a design basis propagates across a sector. Engineering practice is another — and in sectors where regulatory authority is fragmented or consequence-light, it is often the faster one.

Engineering and consulting firms that serve water sector utilities — designing treatment infrastructure, conducting risk and resilience assessments, and advising on capital investment — occupy a position of substantial technical influence over their client base. When a firm adopts an internal design standard, that standard propagates across every project and every client the firm touches. A firm that incorporates the DBA-W Design Basis event categories into its assessment methodology, references DBA-W design principles in its project specifications, and uses DBA-W critical function requirements as the baseline against which it evaluates client systems is, in operational terms, applying a sector-wide design standard — not by regulatory mandate, but through the cumulative force of professional practice across a client portfolio.

The America’s Water Infrastructure Act assessment process is the specific mechanism through which this influence operates most directly. AWIA requires community water systems to conduct risk and resilience assessments, but does not prescribe what an adequate assessment must conclude or what protective measures must result from it. The engineering firm conducting or reviewing a utility’s AWIA assessment is, in practice, defining what adequate looks like for that utility. Firms that bring DBA-W analytical rigor to that process — using DBA-W event categories as the threat envelope, DBA-W consequence thresholds as the inclusion criteria, and DBA-W critical function requirements as the resilience baseline — are filling the gap that the statute left open.

This pathway has historical precedent in analogous sectors. IEEE standards in the electric power sector, American Nuclear Society standards in the nuclear sector, and AWWA standards in the water sector itself all followed the same trajectory: voluntary adoption by leading engineering

firms and consultancies established what the profession considered appropriate practice; that professional consensus created market precedent; regulators and legislators eventually incorporated the standard by reference into binding requirements. The standard did not wait for regulation to give it force — it accumulated force through professional adoption and was formalized afterward. This document is intended to support and accelerate that trajectory for the water sector design basis.

The practical implication for engineering firms is straightforward. Adopting the DBA-W framework as an internal practice standard does not require waiting for a regulatory mandate, sector-wide consensus, or legislative action. It requires treating the design envelope established in Section 4 as the baseline against which client systems are assessed, and treating the critical function requirements as the performance standard against which design and advisory recommendations are made. A firm that does this consistently — across its water-sector client base, in its assessment reports, and in its technical deliverables — is behaving as if the DBA-W framework were a standard. Collectively, firms that make this commitment establish it.

3. Source Material and Analytical Foundation

The water sector possesses a substantial body of risk analysis, threat characterization, and resilience guidance. The Design Basis developed in Section 4 is grounded in that existing work. This section describes the primary source materials and how they are used.

3.1 AWWA J100-21: Risk and Resilience Management of Water and Wastewater Systems

AWWA J100-21 is the sector's most rigorous analytical standard for risk and resilience. It provides a structured methodology for asset characterization, threat assessment, consequence analysis, countermeasure evaluation, and resilience measurement. J100-21 employs a risk-triplet formulation — threat likelihood, vulnerability, and consequence — that is methodologically compatible with design basis thinking.

J100-21 does not itself establish a design basis. It provides the methodology for assessing risk at a system level; it does not define the set of events that all water systems must be designed to withstand. The Design Basis in Section 4 uses J100-21's threat taxonomy and consequence categories as foundational inputs, then takes the additional step of defining the envelope of events that constitutes the sector-wide design basis.

3.2 America's Water Infrastructure Act: 2023 Operational Guide

The 2023 AWIA Operational Guide provides EPA's updated implementation guidance for the risk and resilience assessment and emergency response plan requirements established by AWIA

2018. It identifies the threat categories that assessments must address, the asset categories that must be evaluated, and the documentation standards that emergency response plans must meet.

The Operational Guide's threat taxonomy — covering natural hazards, malevolent acts, and operational failures — provides a starting point for the adverse event categories addressed in the Design Basis. The Guide's emphasis on interdependencies, particularly the relationship between water systems and the electric grid, is directly relevant to the Design Basis's treatment of cascading failure scenarios.

3.3 Water Sector Specific Plan

The Water and Wastewater Systems Sector Specific Plan provides the sector-level risk management framework and an inventory of threat categories recognized at the federal planning level. Its asset categories — treatment facilities, distribution systems, source water, and control systems — map directly onto the Design Basis asset taxonomy. Its threat categories — physical attack, cyber attack, contamination, and natural hazards — are incorporated into the Design Basis event envelope.

3.4 From Risk Assessment to Design Basis: The Analytical Step

Risk assessments characterize events by likelihood and consequence. A design basis defines the envelope of events the system must be demonstrated to withstand, regardless of likelihood. This is a meaningful analytical distinction.

In risk assessment, a low-likelihood, high-consequence event may not drive protective investment if the likelihood is judged sufficiently small. In design basis thinking, the consequence alone — if it reaches a defined severity threshold — places the event within the design envelope regardless of likelihood. Nuclear safety regulation adopted this posture because the consequences of nuclear facility failure are severe enough to warrant protection against events whose probability is very low but whose consequences are unacceptable.

The Design Basis established in Section 4 adopts this posture for the water sector. The criterion for inclusion in the design envelope is consequence severity — the potential for public health impact at population scale, cascading failure across dependent CIKR sectors, or long-duration loss of service to critical facilities — not event likelihood. Events that meet the consequence threshold are in the design envelope. Probability considerations belong to risk management, not to the design basis.

4. The Water Sector Design Basis

The Water Sector Design Basis defines the envelope of adverse events against which water sector infrastructure must be assessed and demonstrated to be resilient. It is organized by event category. For each category, the Design Basis defines the postulated event, the consequence

threshold that places it in the design envelope, and the critical systems whose continued function the Design Basis requires.

4.1 Design Basis Principles

Four principles govern the construction and application of the Water Sector Design Basis.

Consequence-driven inclusion. An event is within the Design Basis if its consequence potential — assuming no protective measures — reaches the threshold of population-scale public health impact, multi-sector cascading failure, or long-duration loss of service to critical dependent facilities. Likelihood is not a criterion for exclusion. Neither is attribution. An event's placement within the Design Basis rests on the reachable path from a control action to its physical consequence, not on the identity, intent, or even the confirmed existence of an adversary. A reachable command that can drive a process to a Design Basis consequence is within the envelope whether it originates with a state actor, an insider, or an operator in error.

This principle is methodologically aligned with the Consequence-driven Cyber-informed Engineering (CCE) methodology developed at Idaho National Laboratory (INL). CCE begins with the identification of the most severe consequences an adversary could achieve against a facility or system and works backward to the cyber attack pathways capable of producing those consequences — explicitly setting aside likelihood as a primary filter in favor of consequence severity. The DBA-W consequence-driven inclusion principle applies the same logic at the sector Design Basis level. If the consequence potential reaches the defined threshold, the event is within the Design Basis regardless of the adversary’s probability of achieving it. Water sector practitioners and engineering firms familiar with CCE will recognize the Design Basis inclusion methodology as operating on the same foundational premise. INL’s CCE program and the DBA-W series share a common analytical orientation; organizations that have conducted CCE assessments will find the consequence threshold determinations that emerge from those assessments directly applicable to Design Basis event category scoping under this framework.

Independence of safety functions. The safety functions identified in the Design Basis — continuous provision of potable water, containment of wastewater, and protection of the public health boundary — must be maintained or restored independently of external support systems when those systems are unavailable. Design Basis compliance requires demonstrating that safety functions can be sustained autonomously for the duration specified in the applicable event category.

Defense in depth. No single protective measure constitutes compliance with the Design Basis. The Design Basis requires layered protective measures such that the failure of any single measure does not produce a Design Basis exceedance. This principle applies to physical, cyber, and operational protective layers. Layered measures are independent only if they do not share a common substrate. Protective layers that depend on a single credential, a single configuration artifact, a single controller program, or a common third-party reference architecture can be

defeated together by one action and do not, in that case, constitute defense in depth. Independence must therefore be demonstrated against a single-substrate defeat — including a shared configuration supplied as a common pattern across many utilities — rather than by counting protective measures (see DBA-MA-DiD). Where independence cannot be demonstrated, what is owed is the disposition step of DBA-MA-DiD §3.4, stated in the adequacy form of its §3.5, which govern and are not restated here. The declared-residual section each facility-class instrument carries is that output form instantiated, and adding redundant barriers does not discharge it. Where a credited function is delivered by more than one train, the trains are credited as independent protection only when they are robust, diverse and independent: two or more trains, differing in kind, and sharing no location, power supply, control system, codebase, vendor or update path, so that no single act defeats them all. Robust and diverse trains that share a room are one train to an adversary. This is the author's ruling of 3 October 2026 on the question DBA-MA-DiD §7.7 holds open, put to Project 6 for that instrument.

Cascading failure as a design basis condition. The water sector's upstream dependencies — principally the electric grid — and its downstream dependencies — hospitals, dialysis centers, food processing, firefighting — are integral to the Design Basis. An adverse event affecting upstream dependencies is a Design Basis event for water systems if its consequence on water system operation reaches the inclusion threshold. Water systems may not claim Design Basis compliance by relying on the continued availability of external dependencies whose failure is itself a Design Basis condition.

4.2 Event Categories

4.2.1 Extended Loss of Electric Power

Postulated event: loss of grid-supplied electric power to a water treatment or distribution system for a duration exceeding the system's emergency generation capacity, or for a duration that depletes consumables required for continued operation.

Consequence threshold: loss of potable water service to a population served, or loss of wastewater containment, for a duration exceeding 72 hours; or loss of service to critical dependent facilities — hospitals, dialysis centers — for any duration.

Required function: treatment and disinfection processes must be maintained or restored within the Design Basis duration using on-site stored energy, emergency generation, or passive means. Wastewater containment must be maintained throughout.

Design basis duration:30 days. This duration is selected to encompass extended grid-restoration scenarios documented in major grid-disruption events and to exceed the infectious-disease incubation window for waterborne pathogens.

4.2.2 Cyber Attack on Operational Technology

Postulated event: a deliberate, targeted intrusion into the SCADA or industrial control systems governing water treatment, disinfection, or distribution, resulting in unauthorized modification of process parameters, loss of operator visibility, or disruption of automated control functions. The postulated event includes modification of the controller program or control logic — not only of process setpoints — and falsification of the indication presented to operators. Either may be effected through the engineering workstation, the vendor engineering software, or the controller project file.

Consequence threshold: unauthorized modification of disinfectant dosing to levels that produce either under-disinfection (pathogen risk) or over-disinfection (chemical exposure risk) in distributed water, or loss of operator control of physical processes for a duration that produces a treatment or containment failure.

Reference event: the February 2021 Oldsmar, Florida, intrusion, in which an unauthorized actor gained remote access to the plant's SCADA system and increased sodium hydroxide concentration to 111 times the normal level. Manual operator detection and intervention prevented a public health event. The Design Basis does not assume operator detection; it requires that the control system architecture preclude unauthorized process modification independent of operator response. Oldsmar is now one of two reference events for this category; the July 2026 coordinated campaign against Minnesota community water systems is the second. Both are characterized in dedicated Reference Event Instruments (DBA-W-RE-1 Minnesota and DBA-W-RE-2 Oldsmar).

Required function: process control systems must be designed such that a cyber intrusion cannot produce a parameter change outside safe operating limits without a positive physical interlock or manual operator confirmation at the local control point. Remote override capability that bypasses physical interlocks is not consistent with Design Basis compliance.

Design basis duration: indefinite. Cyber threats are persistent and do not resolve on a defined timeline. Protective measures must be architectural rather than incident-specific.

4.2.3 Physical Attack on Critical Nodes

Postulated event: deliberate physical damage to a water system's critical nodes — primary treatment facility, major pumping stations, interconnection points with the transmission main, or chemical storage and feed systems — sufficient to disrupt treatment or distribution function.

Consequence threshold: loss of potable water service to the served population, or loss of wastewater containment, for a duration exceeding 24 hours; or chemical release from storage sufficient to produce a public safety hazard in the surrounding area.

Required function: the system must demonstrate that no single node attack produces a consequence exceeding the threshold. Where single-node dependency on a critical function exists, the Design Basis requires either redundancy, rapid restoration capability within 24 hours using pre-staged resources, or both.

Design basis duration:30 days for service restoration to full capacity; 24 hours for restoration to minimum service levels sufficient to supply critical dependent facilities.

4.2.4 Contamination of Source Water or Distribution

Postulated event: introduction of a biological, chemical, or radiological contaminant into source water, a treatment facility, or the distribution system at a concentration sufficient to produce public health consequences if distributed without detection.

Consequence threshold: distribution of contaminated water to any portion of the served population; or detection of contamination after distribution to any critical dependent facility.

Required function: monitoring and detection systems must be capable of identifying contamination events before distribution, or distribution must be capable of immediate cessation upon detection. The Design Basis does not permit reliance on post-distribution detection as the primary protective measure. A distinct adversarial contamination pathway is the deliberate, sustained loss of distribution pressure: a pressure loss that persists, recurs at the adversary's election, or is timed to coincide with other stress can draw untreated ground water into the distribution system through the resulting pressure deficit. The Design Basis treats adversarially induced pressure loss as a contamination initiator, evaluated under conditions where the loss of control persists rather than under the bounded duration assumed for accidental events.

Design basis duration:event-specific. The Design Basis requires that the system demonstrate detection and response capabilities within the time window preceding population-scale exposure, which varies by contaminant type and distribution-system hydraulics.

4.2.5 Extended Loss of Chemical Supply

Postulated event: disruption of the supply chain for treatment chemicals — principally chlorine, sodium hypochlorite, coagulants, and pH adjustment chemicals — for a duration that depletes on-site inventory and prevents continuation of required treatment processes.

Consequence threshold: inability to maintain disinfection residual in the distribution system at regulatory minimums for a duration exceeding 72 hours.

Required function: systems must demonstrate on-site chemical inventory and secure storage sufficient to sustain treatment at full capacity for 30 days without resupply, or demonstrate alternate treatment pathways capable of maintaining regulatory minimums without the disrupted chemical.

Design basis duration:30 days, consistent with the extended-loss-of-electric-power category.

4.2.6 Natural Hazard Exceedance

Postulated event: a natural hazard event — flood, earthquake, hurricane, extended drought — that exceeds the design parameters of one or more critical water system facilities, producing structural damage, source water disruption, or loss of treatment capacity.

Consequence threshold: loss of potable water service to the served population for a duration exceeding 72 hours; or loss of wastewater containment resulting in environmental release.

Required function: systems must demonstrate that natural hazard exceedance of any single facility does not produce a system-wide service failure. Where geographic concentration creates single-point exposure to regional natural hazards, the Design Basis requires either distributed redundancy or pre-established mutual aid agreements with defined capacity commitments.

Design basis duration:30 days for full restoration; 72 hours for minimum service restoration to critical dependent facilities.

4.3 Critical Function Summary

Across all Design Basis event categories, four critical functions must be maintained or restored within the specified durations:

- Continuous provision of potable water meeting Safe Drinking Water Act standards to the served population, with priority service to hospitals, dialysis centers, and fire suppression systems.
- Containment of wastewater to prevent environmental release and public health exposure.
- Maintenance of disinfection residual throughout the distribution system at regulatory minimums.
- Preservation of operator visibility and control of physical processes, including the ability to detect and respond to cyber intrusion, contamination, or equipment failure. Operator visibility must be authentic: the Design Basis requires not only that indication be preserved but that it be true, since manual backup and operator response are worthless against a falsified picture of process state.

These four critical functions correspond directly to the activation thresholds for federal Emergency Support Functions (ESFs) under the National Response Framework. Loss of potable water service to the general population or extended duration activates ESF-3 (Public Works and Engineering), which provides emergency infrastructure assessment, repair, and restoration capability. Loss of potable water service to hospitals, dialysis centers, or other life-safety facilities activates ESF-8 (Public Health and Medical Services), which coordinates patient care continuity, medical facility support, and public health response. Deliberate contamination of source water or the distribution system activates ESF-10 (Oil and Hazardous Materials), which manages hazardous material response and environmental remediation. An extended loss of potable water service to the general population at scale activates ESF-6 (Mass Care, Emergency

Assistance, Temporary Housing, and Human Services), which coordinates emergency water distribution, sheltering, and public notification.

A water system that demonstrates compliance with the Design Basis critical function requirements is simultaneously demonstrating that its architecture is designed to avoid or delay the conditions that require federal ESF activation. The Design Basis is therefore not only a technical standard for water system resilience — it is an instrument for reducing the federal response burden that water sector failures impose. This connection is relevant to FEMA and DHS audiences as well as to EPA and CISA. It should be surfaced explicitly in regulatory and legislative outreach for the DBA-W framework.

5. Cyber as an Integrated Design Basis Requirement

Cyber threats to water sector operational technology are not a future risk category — they are a present and documented condition. The Design Basis treats cyber as an integrated threat running through every event category, not as a separate or subordinate concern. This section establishes the cyber-specific requirements that apply across the Design Basis.

5.1 The OT/IT Boundary

Water sector operational technology — SCADA systems, programmable logic controllers, remote terminal units, historian servers, and human-machine interfaces — evolved from isolated, proprietary systems into networked architectures increasingly connected to enterprise IT networks and, in many cases, to the public internet. This evolution was driven by operational efficiency: remote monitoring, centralized management, and data integration with business systems produce real operational value.

The same connectivity that produces operational efficiency also creates an attack surface. A SCADA system accessible from an enterprise network is accessible from the internet via any compromise of the enterprise network. A remotely accessible human-machine interface that lets operators adjust process parameters from outside the facility also allows any actor with valid credentials — or the ability to obtain or forge them — to do the same.

The Design Basis requirement is not network isolation as an absolute principle — operational connectivity has legitimate value — but a defensible OT/IT boundary that prevents unauthorized physical process modification regardless of the network path by which an attacker might reach the control system.

Three access conditions are named explicitly as Design Basis elements of the boundary. First, shared remote-support and remote-desktop tooling — of the class through which the Oldsmar human-machine interface was reached — is a boundary crossing, and its presence on a control host without individual attribution and multi-factor protection is a Design Basis deficiency. Second, the installed base: control hosts running operating systems or firmware beyond vendor

support, and field devices reached over cellular or other carrier networks, are distinct boundary elements requiring their own isolation and a tracked replacement or decommission path, not instances of general remote access. Third, shared third-party configuration: where an integrator or managed-service provider supplies a common network and control architecture to many unrelated utilities, that shared pattern is a sector-level common substrate, and the obligation to assess and harden it extends to the integrator or service provider, not to the individual utility alone.

5.2 Design Basis Cyber Requirements

The following requirements apply across all Design Basis event categories involving operational technology:

Physical process interlock independence. Chemical dosing systems, pump controls, and valve actuators must have physical interlocks or local manual override capability that functions independently of the networked control system. Unauthorized commands issued through the networked control layer must not be executable at the physical process layer without a positive confirmation step that cannot itself be executed remotely.

Credential architecture. Remote access to operational technology systems must require multi-factor authentication with a hardware element that cannot be replicated through software exploitation. Sharing credentials, default passwords, and vendor remote-access accounts that are not subject to the same authentication requirements as operational accounts is inconsistent with Design Basis compliance. Shared remote-support and remote-desktop applications are subject to the same requirements as any operational remote access; their presence on a control host, unattributed and without a hardware authentication element, is itself a Design Basis deficiency.

Monitoring and anomaly detection. Operational technology networks must be monitored for anomalous behavior — including unauthorized access attempts, unexpected command sequences, and parameter changes outside normal operating ranges — with alerting to operators in real time. Monitoring must be continuous, not periodic.

Incident response and manual backup. Every control function within the Design Basis scope must have a documented manual operating procedure that does not depend on the networked control system. Operators must be trained on manual procedures at intervals sufficient to maintain proficiency. A cyber event that disables the networked control system must not eliminate the operator’s ability to maintain critical functions manually.

Indication integrity. The indication presented to operators — process values, alarms, and the state shown on supervisory displays — must be trustworthy, not merely present. The Design Basis requires that the control architecture preserve the authenticity of indication, because every manual and procedural protection depends on the operator acting from an accurate picture of process state. Manual backup, monitoring, and operator response provide no protection against an adversary who falsifies what the operator sees. Where indication integrity cannot be assured

by architecture, the Design Basis requires an independent means of confirming critical process values that does not share the compromised path.

Protected engineering objects. The engineering workstation, the vendor engineering software, and the controller project file are Design Basis objects and must be protected as such. Modification of the controller program or logic — distinct from modification of a process setpoint — is a named category of unauthorized change that the interlock, monitoring, and recovery requirements must address directly. Access to program-modification capability must be individually attributed, multi-factor protected, and logged, and the integrity of the resident controller program must be verifiable against an offline reference.

5.3 The Oldsmar Lesson

The February 2021 intrusion at the Oldsmar, Florida, water treatment facility is the Design Basis reference event for cyber threats to water-sector operational technology. The technical details are publicly documented: an unauthorized actor gained remote access through a commercially available remote desktop application, operated the human-machine interface in real time, and increased sodium hydroxide concentration in the treatment process to a level that would have caused chemical burns in distributed water.

Two aspects of the Oldsmar event are analytically significant for the Design Basis. First, detection depended entirely on an operator who happened to observe the cursor moving anomalously on a shared workstation. The monitoring and anomaly detection requirements in Section 5.2 address this dependency — detection cannot rely on an operator being present and observant at the moment of intrusion. Second, the control system permitted a remote operator to execute a change to a process parameter without physical confirmation. The physical process interlocks the independence requirement in Section 5.2, which addresses this directly.

The Oldsmar event did not produce a public health consequence. That outcome was a function of chance, not of protective architecture. The Design Basis is constructed to ensure that the absence of a public health consequence does not depend on chance.

Two further points refine the Oldsmar lesson. First, at Oldsmar the indication presented to the operator was true: the operator saw an accurate picture of the setpoint change and acted on it. The lesson the Design Basis drew therefore reached detection independence but not indication integrity, because in the source event indication was never at issue. The July 2026 Minnesota campaign, in which indication was falsified, is the reference event that exposes the difference, and the indication-integrity requirement in Section 5.2 closes it. Second, the public attribution of the Oldsmar event has since been contested: the federal investigation did not confirm a targeted intrusion, and an official later characterized the incident as a possible operator error. This does not weaken its standing as a reference event. The Design Basis rests on the reachable action-to-consequence path (Section 4.1), which is unchanged whether the setpoint change originated with

an intruder, an insider, or an operator in error. Both events are characterized in full in Reference Event Instruments DBA-W-RE-1 and DBA-W-RE-2.

5.4 Machine Learning and Generative AI in Water Sector Control Systems

The term artificial intelligence (AI) encompasses a broad family of computational techniques. For Design Basis purposes, two subcategories require distinct treatment. Machine learning (ML) encompasses algorithms that learn from data to improve performance — anomaly detection, predictive maintenance, pattern recognition — and can be deterministic: given bounded inputs, ML algorithms produce bounded, testable outputs verifiable through semi-formal methods. Generative AI (Gen AI) encompasses large language models and similar systems that produce novel outputs from learned distributions. Gen AI is non-deterministic: the same input can produce different outputs across execution cycles, and formal verification methods do not apply. Whether the requirements of Section 5.5 reach a component does not turn on this distinction, however. It turns on whether the component meets the determinism carve-out of FD-LD §4.3, which governs scope: Gen AI is the principal case within scope, and an ML component that is not deterministic within a stated input bound and formally testable as deployed is within scope with it. Where this document uses ‘ML and Gen AI,’ it refers to these two analytically distinct subcategories, not to AI as an undifferentiated whole.

ML and Gen AI are entering water sector operations through two distinct pathways: vendor-embedded ML and Gen AI components in commercial control system products, and operator-facing Gen AI decision-support tools such as large language model assistants deployed on enterprise IT networks. Both pathways introduce risk categories that the Design Basis must address explicitly.

Vendor-embedded ML and Gen AI represent the more immediately consequential pathway because these components are designed to reside within the OT boundary. Vendors of SCADA platforms, human-machine interfaces, and distributed control systems are incorporating ML and Gen AI into their standard product offerings, in some cases with capabilities described as autonomous process optimization. Operator-facing Gen AI tools — including commercially available Gen AI assistants and utility-developed decision-support applications — are deployed on enterprise IT networks and increasingly integrated into operational workflows that influence physical process decisions.

The risk profile of these two pathways differs in kind, not only in degree. Vendor-embedded components with command authority over physical processes present a direct Design Basis risk: unauthorized or erroneous process parameter changes can occur through the ML or Gen AI component without operator confirmation. Operator-facing Gen AI presents a different risk. Where its output would itself make a change once accepted — a recommended setpoint an operator enters — the recommendation and its acceptance are one action, placed on the consequence of the change (FD-BL §4.3). Where its output would make no change of its own — a summary, a diagnosis, a synthesis of plant state — the operator who then decides is taking his

own action, and his reliance on the output does not move the obligation onto it (FD-BL §4.2; FD-BL-D1 §7). The hazard in that second case is not command authority but a corrupted basis for authorization: an operator who treats Gen AI output as authoritative and acts on it without independent confirmation is confident and wrong, and the path from the output to the consequence is invisible to monitoring systems. That output carries the indication-integrity obligation of FD-BL-D2.

The Design Basis requirement for both pathways is architectural enforcement of a clear boundary between the advisory function of in-scope components and physical command execution, together with the indication-integrity obligation of FD-BL-D2 wherever advisory output can ground an authorization. That boundary and its enforcement mechanism are established in Section 5.5.

5.5 The Command-Broker Architecture and the Bright Line

The command-broker architecture is the Design Basis requirement for every component whose outputs can affect physical process parameters in water-sector control systems and that does not meet the determinism carve-out; the scope rule is FD-LD §4.3, which governs and is not restated here. It is the Gate: the architecture incapable of executing an above-line action without authorization, sitting between the advisory layer and the physical command layer. The Bright Line is not that boundary. Placement is governed by FD-BL, on the consequence of the individual action, assessed per action; this framework applies that definition and does not vary it. For the water sector this framework adopts a conservative implementation: every action crossing from the advisory layer into the physical command layer is treated as above the line unless a documented per-action consequence assessment places it below. The layer boundary is the enforcement surface, not the placement criterion. Nothing crosses the Bright Line without passing through a human command broker, and the boundary is enforced by architecture, not by procedure or operator discipline.

Governing instruments. This framework does not own the concepts it applies. The Bright Line — its placement criterion, granularity, scope of application, and the obligation it creates — is defined in FD-BL, which governs. The Broker role that discharges that obligation is defined in FD-BR. Layer decomposition, the determinism boundary as a scope gate, and the machine-learning carve-out are defined in FD-LD. Two determinations bind under FD-BL: FD-BL-D1 on the mode of discharge and FD-BL-D2 on indication integrity. FD-EV reserves the term “envelope”. Instruments of the Foundational Definitions family are cited here by designation and without a version token, under the citation-form determination; FD-REG, the Governing Stack Register, is authoritative for their version state and resolves each designation to the issue in force. Propagation is by reference to those instruments, not by restating their text.

The canonical cross-sector instrument governing the command-broker architecture is the Command Broker Framework, Cross-Sector (CB-Framework_Cross_Sector), published by Eclectic Technologies and cited by designation; CB-REG, the Command Broker Family

Register, is authoritative for its version state. That document establishes the broker concept, the Bright Line principle, the three-tier criticality framework, and the distinction between the Command Broker and the Influence Broker at the cross-sector level. It is the authoritative source for all requirements that this section implements in the water sector context. The CB-Framework, in turn, draws its criticality bands and the Bright Line concept from Generative AI Risk in Industrial Control Systems: A Study for Critical Infrastructure Operators, Regulators, and Suppliers (Version 29), which supersedes the earlier ISAGCA-attributed "A Study of AI Risks in Critical Infrastructure — Vendor Agnostic" (Version 28).

This section is the water sector implementation of the CB-Framework. Where this section specifies water-sector-specific consequence thresholds, application classifications, or compliance requirements, those specifications apply within the CB-Framework's architectural constraints. Where this section is silent on a requirement that the CB-Framework addresses, the CB-Framework governs.

5.5.1 The Three-Tier Criticality Framework

The CB-Framework establishes a three-tier criticality framework, derived from that study, that assigns ICS applications to tiers based on the severity of consequences. Each tier defines permissible AI behavior in the water sector context as follows:

Highest Criticality — no autonomous Gen AI action. Applications at the highest criticality tier are those whose malfunction, misoperation, or intentional misuse could produce the consequence thresholds defined in Section 4: population-scale public health impact, loss of wastewater containment, or loss of service to critical dependent facilities. At this tier, Gen AI may guide and inform the operator through an interactive interface, but no autonomous Gen AI action is permissible. A human command broker must be in the control loop for every change to a physical process parameter. A component that is deterministic within a stated input bound and formally testable as deployed meets the carve-out and is outside the Command Broker obligation for its in-bound actions (FD-LD §4.3). The carve-out is not a grant of ungoverned autonomy, and it is not a finding that those actions are below the Bright Line; §5.5.3 states the distinction and it is not restated here.[1]

Mid-Level Criticality — Gen AI informs, human decides. Applications at the mid-level criticality tier involve moderate consequences that do not rise to the Design Basis threshold. Gen AI may inform operators of potential actions; operators then decide which action to execute. A human remains in the command loop. A component meeting the determinism carve-out is outside the Command Broker obligation for its in-bound actions on the same terms as at the highest tier (FD-LD §4.3).

Lowest Criticality — the tier lightens assessment; it does not license autonomy. Applications at the lowest criticality tier involve consequences that do not affect critical functions or safety. The lightest assessment, qualification, and audit regime applies, and Gen AI may serve as a virtual assistant. Whether any action it takes may proceed without a command broker is decided

by that action's placement on its own consequence and not by the tier: band assignment is not placement, and a lowest-tier application may contain above-line actions, each of which requires a command broker (FD-BL §4.3). A component meeting the determinism carve-out is outside the Command Broker obligation for its in-bound actions on the same terms as at the highest tier (FD-LD §4.3).

For the water sector Design Basis, the highest-criticality tier governs all applications that control or influence disinfection dosing, treatment process parameters, distribution system pressure and flow management, and wastewater containment. These are the applications in which above-line actions are to be expected. The tier fixes the rigor of their assessment, qualification, and audit; it does not place their actions, each of which is placed on its own consequence (FD-BL §4.3).

5.5.2 The Command Broker

The command broker is the human operator who decides — who brokers — a course of action based on process observables and advisory output. The command broker function exists to ensure that no recommendation from a component within scope of FD-LD §4.3, however well-grounded in data, produces a change in a physical process parameter without a human decision interposed between the advisory output and the physical action.

The command broker is not a procedural role that can be satisfied by policy language or training alone. The command-broker architecture requires that the control system enforce the broker function mechanically: the system must be incapable of executing an above-line action originating from a component within scope of FD-LD §4.3 — under this framework's conservative implementation, any physical process command that a documented per-action assessment has not placed below the line (§5.5) — without a positive human confirmation step that is local, auditable, and not itself executable remotely or by another software component.

This requirement applies regardless of how a component is characterized by its vendor (FD-BL §4.2). A vendor that describes their embedded Gen AI as providing "autonomous process optimization" or "direct control integration" is describing a capability that, in the highest-criticality water sector applications, is inconsistent with Design Basis compliance. The vendor characterization does not govern; the Design Basis requirement governs. Asset owners and operators are responsible for verifying that every in-scope component deployed in their highest-criticality applications enforces the command-broker boundary, and that no output of such a component reaches physical command execution without human interposition unless a documented per-action assessment has placed that action below the line.

The qualification requirements for individuals appointed to the command broker role — competency domains, assessment methodology, organizational structure requirements, and audit obligations — are established in the CB-IB Qualification and Onboarding Standard (CB-IB_Qualification_Standard). That document is the implementing instrument for CB qualification across all sectors. Water sector operators must ensure that individuals performing the command broker function in the highest-criticality applications are qualified under that standard. The water

sector annex to the Qualification Standard, when drafted, will specify sector-specific competency requirements and stress scenarios appropriate to water sector operations.

5.5.3 The Bright Line Applied to Vendor-Embedded ML and Gen AI

Vendor-embedded ML and Gen AI in water sector control systems must be assessed against the Bright Line before deployment in any application at the highest-criticality tier. The assessment must address three questions:

Does the ML or Gen AI component have any pathway to physical command execution without human confirmation? If yes, that pathway must be eliminated or rendered non-functional before the product is deployed in a highest-criticality application. The vendor's assurance that the pathway is "safe" or "controlled" is not a substitute for architectural enforcement. The question binds a component within scope under the second question; a component that meets the carve-out is governed instead in the ecosystems FD-LD §4.3 names.

Is the component deterministic and formally testable? Determinism decides scope, not placement (FD-LD §4.3): a component that is deterministic within a stated input bound and formally testable in its deployed configuration meets the carve-out and is outside the Command Broker obligation for its in-bound actions. The carve-out is not a grant of ungoverned autonomy (FD-LD §4.3), and it does not follow that the component's actions are "below the line"; placement is a property of actions and is set by consequence (FD-BL §4.2). The input bound must be stated and verified as part of the assessment, and an action produced while the component's inputs lie outside it is not covered. Gen AI components, including large language models and other non-deterministic AI systems, do not meet the carve-out and remain within scope, as does any component never formally tested as deployed. Each of their actions is placed on its own consequence, and at the highest-criticality tier they may not act autonomously (§5.5.1), whatever vendors claim about their reliability or safety.

Has the Gen AI component been subjected to red-teaming? Red-teaming of Gen AI systems — simulating real-world adversarial manipulations, including prompt injection, training data poisoning, and adversarial inputs designed to produce unsafe outputs — is a required element of Design Basis compliance for any Gen AI component deployed in a water sector control system. Red-teaming must be conducted by parties independent of the vendor and address water-sector-specific attack scenarios.

5.5.4 The Bright Line Applied to Operator-Facing Decision Support

Gen AI decision-support tools deployed on enterprise IT networks — including commercially available large language model assistants, utility-developed operational Gen AI, and tools such as Microsoft Copilot configured for operational use — are within scope of Bright Line analysis (FD-LD §4.3), and what the obligation requires depends on what each output would do. Output that, once acted upon, would itself make a change to a process parameter — a recommended dosing adjustment, or a control-system entry the tool pre-populates — is an action: the

recommendation and the operator's acceptance of it are one action, placed on the consequence of the change (FD-BL §4.2, §4.3), and where that consequence is above the line the acceptance is the Command Broker's authorization and must meet the obligation. Output that would make no change of its own — a summary, a diagnosis, a synthesis of plant state — is placed below the line as a process action, and an operator who then decides on a change takes that action himself; reliance on the output is not influence that moves the obligation onto it (FD-BL-D1 §7). Where such output can function as the basis for an above-line authorization, it carries the indication-integrity obligation of FD-BL-D2. Neither outcome turns on the application's criticality tier.

The obligation reaches the workflow, not only the system architecture. A decision-support tool that recommends a disinfection dosing adjustment, and whose recommendation an operator implements by entering the adjustment directly into the control system, has proposed an action that the operator's entry accepts; the recommendation and its acceptance are one action, placed on the consequence of the dosing change, and for disinfection dosing that consequence is above the line. The authorization is substantive only if it does not rest on the tool's output alone. Under FD-BL-D2 I3 that output may not be the sole basis for an above-line authorization: the operator must have an independent path by which to confirm the process state before accepting — not simply act on the recommendation — and where no such path exists the authorization is unavailable on that basis. Where uncorroborated output of the tool is an input on which placement depends, the action it bears on does not proceed without in-the-moment Command Broker authorization, and no placement changes (FD-BL-D2 §4.3). Workflow design must enforce these conditions. Tools that present Gen AI recommendations as actionable defaults or pre-populate control system inputs from Gen AI output turn acceptance into a formality and defeat the substantive authorization FD-BR §4.5 requires, whether the tool is classified as a decision-support system or as a control system.

Additionally, cloud-connected Gen AI tools pose a network boundary risk independent of the command-broker concern. If a decision-support tool deployed on the enterprise IT network has any pathway — direct or through integration — that reaches the OT network, the OT/IT boundary established in Section 5.1 is compromised. The network boundary and the command-broker architecture are independent requirements; both must be satisfied.

5.5.5 Model Integrity and Supply Chain

Gen AI models are updated through vendor channels — model weights, training data, and fine-tuning — on schedules and through mechanisms that are typically outside the asset owner's direct control. This creates a supply chain attack surface with no analog in traditional OT cybersecurity. A compromised model update does not trigger OT intrusion detection systems. It does not appear as a network anomaly. It appears as a software update from a trusted vendor.

Design Basis compliance requires that asset owners and operators establish model provenance requirements for any Gen AI component deployed in their systems: documented verification of training data sources, version control with change logging for model updates, vendor

certification of update integrity, and the ability to roll back to a prior model version if anomalous behavior is detected. These requirements apply to both vendor-embedded Gen AI and operator-facing decision-support tools. The absence of industry-standard frameworks for Gen AI model supply chain assurance in the water sector is a recognized gap that the tiered framework — particularly the L2 and L3 documents — will address in greater detail.

6. Tiered Framework Architecture

The Design Basis established in this document is a sector-wide standard — a definition of what water sector infrastructure must be capable of withstanding. It does not, by itself, establish the graduated operational response framework that facilities need to manage escalating conditions, or the monitoring and data-sharing infrastructure required to detect the development of Design Basis conditions in real time. Those functions are addressed in subsequent DBA-W series documents organized as a tiered framework.

The tiered framework architecture mirrors the structure proven in the nuclear sector under the DBA-MA series. Four instrument levels address distinct analytical and operational functions:

L1 — National Framework (this document). Establishes the sector Design Basis. Defines the design envelope, critical functions, and design basis principles. This is the standard against which compliance is measured.

L2 — Formal Adverse Event Description. Develops each Design Basis event category into a detailed scenario description with consequence chain analysis. Analogous to the Accident Description function in the nuclear DBA-MA L2. Establishes the analytical basis for facility-specific assessment. The L2 will develop the command-broker architecture and Bright Line requirements of Section 5.5 in the context of each Design Basis event category, with particular attention to the cyber-attack-on-OT scenario.

L3 — Trigger and Escalation Framework. Defines observable indicators that govern the transition of a water system from normal operations through elevated readiness to emergency response posture. Establishes decision authority, action thresholds, and coordination requirements.

L4 — Monitoring and Data-Sharing Guidance. Provides implementation guidance for the monitoring sources and information-sharing relationships required to detect Design Basis conditions as they develop in real time. Establishes the information infrastructure on which the L3 Trigger Framework depends.

ISA/IEC 62443, the industrial control system security standard series, is incorporated by reference within the DBA-W framework as the governing standard for OT/IT security architecture. It is positioned alongside the DBA-W tiered framework — not within it. DBA-W is the governing instrument. ISA/IEC 62443 provides the technical security architecture standards that DBA-W compliance requires for cyber-related design basis elements.

7. The Military Action Extension

7.1 MA Extension Applicability Determination

The Military Action extension of the Design Basis does not apply uniformly to every water sector facility. Application of the MA extension is a mandatory facility-level determination governed by a two-factor test established at the cross-sector level in the DBA Series Gaps and Recommendations document (§1.7) and implemented here for the water sector. A facility that completes this determination and documents a reasoned basis for deferral is not required to engage with the remainder of this section. A deferred determination is not permanent; it must be revisited on the cycle defined in Section 7.1.3 and re-triggered immediately upon the geopolitical indicators identified there.

7.1.1 Primary Factor: Geolocation

The primary factor is geolocation: the facility's physical location relative to credible military threat actors, contested or militarized borders, active or latent conflict zones, and the proximity of allied and adversary military assets. A water sector facility in a geopolitical environment that presents a credible, near-term military threat to its host nation's infrastructure is presumed to require the MA extension. A facility in a stable interior jurisdiction with no credible near-term military threat to its host nation may document a geolocation-based deferral, provided the deferral is supported by a written threat assessment referencing the current geopolitical context and updated on the cycle defined in Section 7.1.3. Geolocation alone does not determine applicability; the secondary factor must also be assessed. A facility in a low-risk geolocation may nonetheless require the MA extension if it scores as high strategic value under Section 7.1.2.

7.1.2 Secondary Factor: Strategic Value

The secondary factor is strategic value: whether the facility's loss or degradation produces a consequence chain that a state actor would find strategically valuable as a military objective. Strategic value creates targeting incentive independent of geographic proximity. A water sector facility meets the strategic value threshold under any of the following four indicators.

Sole-source supply. The facility is the only source of potable water for a defined population, whether due to the absence of alternative infrastructure or to logistical constraints that preclude delivery of an alternative supply within the Design Basis duration window established in Section 4.

Military or defense-critical installation dependency. The facility supplies, or is the primary backup supply for, a military installation, defense industrial facility, or continuity-of-government facility.

Critical dependent facility concentration. The facility supplies a hospital, dialysis center, or equivalent life-safety facility cluster whose patient population cannot be safely transferred or sustained without continuous water service.

Industrial cascade. Loss of the facility's water supply would trigger a consequential failure in a co-located or directly dependent industrial process — nuclear plant cooling, chemical manufacturing, data center thermal management, or equivalent — whose own consequence chain meets or exceeds the water sector Design Basis threshold.

7.1.3 Time-Varying Property and Re-Determination Triggers

The MA extension applicability determination is not permanent. Geopolitical stability is not a fixed characteristic of a site or region across a facility's operating life. A facility that documents a reasoned deferral must re-evaluate that determination on a five-year cycle coincident with the facility's AWIA risk and resilience assessment review obligation, or more frequently if the facility's threat environment changes materially. Re-determination must be triggered immediately — outside the five-year cycle — upon any of the following geopolitical indicators: massing of hostile forces within operational striking distance of the facility's host nation; escalation of a regional conflict to a level that could involve the facility's host nation or directly dependent military assets; explicit public or credible non-public threats against water sector infrastructure by a state or state-sponsored actor; or a confirmed attack on a water sector facility of the same facility class in a geopolitically comparable jurisdiction. A deferral that was valid at the last assessment is not valid following a triggering indicator until re-determination is completed and documented.

7.2 Why the Non-Adversarial Design Basis Is Insufficient

The Design Basis established in Sections 4 and 5 is necessary but insufficient as a complete analytical framework for the water sector. Each bounding event in Section 4 shares a characteristic that distinguishes it from the military-action threat tier: it is not chosen. Equipment fails according to probabilistic distributions. Hurricanes follow meteorological paths. Earthquakes occur along fault lines. The Oldsmar intrusion was executed by a financially motivated actor using a commercially available remote-access tool — criminal opportunism, not state strategy. The system designer can analyze historical frequencies, model consequence chains, and invest in protection against the events most likely to occur within a defined planning horizon. The adversary does not operate within this framework.

Military action against water infrastructure introduces four characteristics that are absent from the non-adversarial Design Basis and that require an extended analytical framework to address. These four characteristics were identified in the DBA-MA parent framework in the nuclear context. They apply without modification to the water sector — and in some respects apply with greater force, because the water sector's consequence chain reaches public health directly, rapidly, and in ways that are largely invisible until they produce casualties.

7.3 The Four Military Action Characteristics

7.3.1 Intentionality

The adversary chooses its targets. It does not attack randomly or according to a probabilistic distribution. It studies the sector, identifies the highest-consequence targets, assesses the defensive posture of those targets, and selects the attack configuration most likely to achieve the desired strategic effect. For the water sector, intentionality means the adversary will not attack in summer when an attack timed to the winter peak demand for heating fuel and firefighting capacity compounds the consequences. It means the adversary will target the water treatment facility serving the hospital cluster, not the smaller system serving the residential area. It means the adversary will attack the sodium hypochlorite storage and feed system — the component whose loss most rapidly propagates to a disinfection-residual failure in the distribution system — not the pump, which has a three-day spare-parts lead time.

The implication for the Design Basis is that the non-adversarial bounding events assume the worst event is determined by physics and probability. The military-action bounding event is determined by adversary choice. The water system must be hardened not only against the probabilistically most likely events but against the strategically most consequential targets, as assessed by a sophisticated adversary who has studied the system's architecture.

7.3.2 Persistence

Military action does not affect the water system's recovery timeline. A pipe rupture is repaired when crews and materials are available. A pump failure is corrected when the replacement part arrives. Military action can be sustained: an adversary that understands the water system's recovery architecture can plan follow-on actions timed to defeat the recovery. The portable generation deployed to sustain treatment through a grid outage is itself a target. The chemical resupply convoy is a target. The mutual-aid crews mobilizing from neighboring utilities are operating in a threat environment that the non-adversarial bounding events do not contemplate.

Persistence is particularly consequential for the water sector because the system's most critical protective function — maintaining disinfection residual in the distribution system — degrades continuously in the absence of treatment. Unlike an electrical outage, which can be tolerated for some duration before life-safety consequences develop, a loss of disinfection residual begins creating pathogen risk immediately and invisibly. An adversary who understands this dynamic will target the treatment function and then prevent restoration long enough for the distribution system to become a public health hazard.

7.3.3 Precision

Modern state actors can execute precision attacks against specific system components. The adversary does not merely threaten the water system; it can target the supervisory control system,

whose compromise would allow manipulation of chemical dosing parameters — replicating the Oldsmar scenario with state-actor pre-positioning, no operator detection, and coordinated physical action that denies the manual intervention that prevented the Oldsmar event from causing casualties. Precision targeting allows the adversary to achieve maximum public health impact while minimizing the attack's signature, complicating both attribution and response.

In the cyber dimension, precision means the adversary does not conduct a broad network intrusion — it targets the specific SCADA historian that stores dosing setpoints, or the specific remote terminal unit that controls the sodium hypochlorite feed pump, or the specific human-machine interface whose compromise allows setpoint modification. The Command-Broker architecture established in Section 5.5 is the primary protective measure against precision cyberattacks. Its adequacy under military-action conditions — where the attacker has had months of pre-positioned access to understand the architecture before executing — must be assessed explicitly.

7.3.4 Denial

Military action can deny site access. The operators who would detect and manually correct an anomalous dosing setpoint cannot access a facility under active military threat. The inspection and sampling teams that would verify distribution system water quality cannot operate safely in a conflict zone. The repair crews tasked with restoring damaged treatment infrastructure cannot reach the site.

The local sovereignty principle — established in the DBA-MA parent framework and inherited here — is the engineering response to denial: protective systems must be capable of operating passively, requiring no external power, communications, or operator action to initiate. For the water sector, this principle extends beyond safety system design into the treatment process itself: chemical dosing systems must fail to a safe condition — either maintaining the last valid setpoint or defaulting to a conservative dosing rate — when operator access is denied. A treatment system that requires continuous operator presence to maintain safe operation is not consistent with Design Basis compliance under military-action conditions.

7.4 The Oldsmar Bridge: From Non-State Event to Military-Action Design Basis

The Oldsmar incident is the reference event for the non-adversarial cyber bounding event in Section 4.2.2. It is also the analytical bridge to the military-action design basis. The contrast between the two scenarios reveals exactly what the military-action extension adds.

The Oldsmar event: a financially motivated actor using a commercially available remote-desktop application gained access to an unprotected HMI, manipulated a chemical dosing setpoint to a lethal level, and was detected by an operator who happened to observe the cursor moving on a

shared workstation. Manual intervention prevented a public health event. The outcome was a function of chance.

The military-action analog: a Tier 1 state actor has spent six months pre-positioning within the water system's SCADA architecture. It has mapped every dosing setpoint, every alarm threshold, and every operator override pathway. It executes the dosing manipulation during a winter storm that has degraded grid power to the facility and strained the utility's operational staffing. It simultaneously activates a denial-of-service condition on the utility's SCADA communications to the operations center. It times the attack for the weekend night shift, when the minimum staffing complement is on duty and supervisor response time is longest. It does not move the cursor visibly on the shared workstation — it manipulates the setpoint through a process historian interface that requires no HMI interaction. There is no anomalous cursor to observe. The operator's manual detection pathway — the one that saved Oldsmar — does not exist in this scenario.

The architectural requirement is the same in both scenarios: the control system must preclude unauthorized process modification independent of operator detection. But the adversary's capability in the military-action scenario requires that the architecture be validated not only against the Oldsmar-class attacker but against a state actor who has studied the architecture, identified its weakest point, and designed an attack to defeat every detection mechanism. That validation standard is the military action extension of the cyber Design Basis. The military-action scenario above is a constructed analog — an account of what a capable, directed state actor could do through the same class of access — and not a claim that the Oldsmar event was itself a state attack. Its attribution is in fact contested (Section 5.3). The analog's force does not depend on that question; it derives from the reachable capability, not from the identity of the Oldsmar actor.

7.5 DBA-MA Threat Framework for the Water Sector

7.5.1 Threat Actor Taxonomy

Tier 1 — State Actor. Nation-states with advanced persistent threat capability and the demonstrated willingness to employ combined cyber, kinetic, and influence operations against critical infrastructure. Russia has demonstrated sustained attacks on Ukrainian water and energy infrastructure. China has documented pre-positioning activity in U.S. critical infrastructure networks, including water systems. Iran has employed cyber operations against water sector SCADA systems in the United States. Tier 1 actors have the intelligence resources to conduct multi-year pre-positioning campaigns, the technical capability to execute precision cyber-kinetic attacks, and the strategic patience to time attacks for maximum public health consequences.

Tier 2 — Near-State Proxy. Paramilitary organizations and hybrid warfare contractors operating under state direction. Tier 2 actors provide plausible deniability for kinetic operations. The relevant scenario involves a Tier 1 state actor directing a Tier 2 proxy to execute physical attacks

on water treatment or distribution infrastructure while the Tier 1 actor maintains the SCADA compromise that amplifies the physical attack's consequences.

Tier 3 — Sophisticated Non-State Actor. Criminal organizations, hacktivists, and ideologically motivated actors. The Oldsmar incident is a Tier 3 event. Tier 3 threats are addressed by the non-adversarial Design Basis cyber requirements in Section 5. The military-action extension addresses Tier 1 and Tier 2. Two qualifications apply. First, the attribution of the Oldsmar event is contested, and it is more accurately recorded as unattributed; the tier scale should not be read as asserting a confirmed actor class for it. Second, tier denotes direction — who is acting and to what end — not a ceiling on capability. Offensive capability developed by Tier 1 actors proliferates, so a Tier 3 actor may wield state-grade tools; applicability of the extension turns on the capability an actor can bring to bear, assessed separately from the actor's tier.

7.5.2 Attack Modality Taxonomy

Kinetic Direct. Physical destruction of water infrastructure assets using explosives, incendiary devices, or precision munitions. Treatment plant process equipment, chemical storage and feed systems, major pumping stations, and water tower and reservoir infrastructure are the primary kinetic targets. The consequence of a well-executed kinetic attack against a treatment plant's chemical system is immediate: loss of disinfection capability within hours, depletion of distribution-system residuals within days, and pathogen risk within the timeframe dictated by the system's hydraulic residence time.

Cyber-Kinetic Combined Arms. Integration of SCADA compromises with kinetic action. The cyber component serves to suppress alarms and protective systems before the kinetic strike, manipulate process conditions to amplify the consequence of physical damage, delay detection and attribution, and complicate recovery by corrupting control system state. This is the military-action analog of the Oldsmar scenario — with kinetic execution added and detection pathways systematically closed.

Contamination as a Weapon. Deliberate introduction of biological, chemical, or radiological agents into source water, treatment processes, or the distribution system. This modality exploits the water sector's unique consequence pathway: contaminated water distributed to consumers produces public health consequences that are geographically broad, temporally delayed from the point of introduction, and extremely difficult to attribute rapidly. The Design Basis contamination event category in Section 4.2.4 addresses this modality; the military-action extension requires that the protective architecture be validated against a state actor who has studied the system's monitoring and detection capabilities and designed the contamination to defeat them.

Supply Chain Denial. Interdiction of the supply chains for critical treatment chemicals, replacement equipment for pumping and treatment infrastructure, and emergency generation fuel. A Tier 1 adversary that understands the water sector's dependence on chlorine and sodium hypochlorite supplies and the limited on-site chemical inventory at most utilities can plan supply

chain interdiction operations that extend disruption well beyond the period of physical damage. The 30-day chemical inventory requirement in Section 4.2.5 is the Design Basis response to non-adversarial supply chain disruption; the military-action extension requires assessing whether that inventory is adequate when the adversary is actively interdicting resupply.

7.6 The Extended Design Basis: DBA-MA for Water

The DBA-MA extension does not replace the non-adversarial Design Basis — it builds on it. Each non-adversarial bounding event has a military-action analog: the same initiating condition, deliberately caused, precisely targeted, strategically timed, and executed to defeat the recovery architecture. The extended Design Basis asks: given that the adversary has studied this water system, identified the highest-consequence targets, and can execute a precision combined cyber-kinetic attack, what is the bounding military-action event, and what does adequate protection against that event require?

The DBA-MA bounding event is in every case more severe than its non-adversarial analog, for four reasons that follow directly from the four characteristics above. First, the adversary selects the highest-consequence configuration; the non-adversarial event occurs according to probability. Second, the adversary times the attack to maximize consequence; the non-adversarial event is timing-indifferent. Third, the adversary can coordinate simultaneous events that defeat the water system's single-scenario resilience mechanisms; the non-adversarial events are independent. Fourth, the adversary can deny the access on which recovery depends; the non-adversarial recovery framework assumes unimpeded access.

The Command-Broker architecture and Bright Line requirements established in Section 5.5 are the primary cyber-defensive measures against the military-action threat. Their design must be validated not only against the opportunistic attacker who was stopped at Oldsmar, but against a state actor who has had months to study the architecture and design an attack to defeat every human detection pathway. That validation — including red-teaming by parties independent of the vendor and with water-sector-specific knowledge of attack scenarios — is the military-action extension of the Design Basis cyber compliance standard.

8. Consequence Interaction and Cross-Sector Dependencies

8.1 The Water-Electricity Interdependency

The water sector's most consequential upstream dependency is the electric grid. Water treatment and distribution are energy-intensive: treatment processes require continuous power for chemical feed, filtration, ultraviolet disinfection, and blending; distribution requires power for pumping throughout the pressure zones; wastewater treatment requires power for aeration, biological treatment, and effluent pumping. A grid reliability event that degrades power to a water utility

triggers a chain of consequences whose endpoint — loss of treatment function and distribution system pressure — produces public health impacts within days.

The interdependency runs in both directions. Power generation requires cooling water; thermoelectric generation facilities — nuclear, natural gas, and coal — withdraw large volumes of water for once-through or recirculating cooling. A sustained water supply disruption that affects a power generation facility's cooling water intake can force the facility offline, compounding the grid reliability event that may have initiated the water supply disruption in the first place. Under military-action conditions, an adversary who understands this interdependency will attack the seam between the two sectors — the water systems that serve power generation facilities, or the power infrastructure that sustains water treatment — to initiate a self-reinforcing cascade. The reverse direction carries an obligation this series owes. Under DBA-EN-SF §5.4, a water facility that is the sole cooling-water supply path for a thermoelectric generation facility carrying a Tier 2 or Tier 3 consequence designation in the DBA-ES series must carry tightened acceptance criteria for that supply relationship in the instrument governing its facility class; each of FC1 through FC5 carries that requirement conditionally.

8.2 Life-Safety Cascade: Hospitals and Dialysis Centers

The water sector's most acute downstream dependency is the life-safety infrastructure — hospitals, dialysis centers, long-term care facilities — that depends on a continuous potable water supply for patient care. Dialysis is the most time-critical dependency: hemodialysis machines require large volumes of highly purified water, and dialysis centers cannot operate from stored water for more than a few hours. A water supply disruption affecting a dialysis center requires immediate patient transfer to alternative facilities. This logistically complex operation becomes life-threatening if alternative capacity is unavailable or if the disruption is widespread.

The Design Basis duration requirements in Section 4 — 72 hours for general service restoration, with priority service to critical dependent facilities for any duration — are driven primarily by this cascade. Under military-action conditions, the adversary who understands the dialysis center's dependency will time an attack to exploit it: a winter storm that limits patient transfer options, a simultaneous attack on alternative water supply routes, or a timing that coincides with a regional emergency that has already strained hospital surge capacity. The adequacy of the Design Basis duration requirements must be assessed against the adversary's ability to exploit these compounding factors.

8.3 Fire Suppression Dependency

Municipal fire suppression depends on the water distribution system pressure and flow. A sustained loss of distribution system pressure — from a treatment plant outage, a major main break, or a pumping station failure — degrades fire suppression capability across the affected

service area. Under military-action conditions, this dependency creates a second-order consequence: if the attack on water infrastructure coincides with or triggers fires — as a kinetic attack on a treatment plant's process equipment would — the water system failure simultaneously creates fires and degrades the suppression capability needed to fight them. The adversary who understands this dynamic can design an attack that exploits both consequences simultaneously.

On the author's decision of 6 October 2026, this dependency carries an elevation rule, stated jointly with the Emergency Services series. A fire district that has no alternate supply of its own — no tankers, drafting sites, static tanks or relay capability sized to the fires it must be designed for — depends on the water system as its only means of fighting fire. A water facility that is the only supply for such a district carries a tightened acceptance criterion for the pressure and flow it delivers to that district, on the same structure as the elevation the series applies for hospitals and dialysis centers. A district that holds an alternate supply does not elevate the water facility, and the alternate is the district's to hold (*Response Forces and Stations* §4.5, §8.3). The criterion is not yet quantified, and who bears its cost is the question Project 9 OI-17 already asks for cooling water.

8.4 Seasonal Amplification

The consequence envelope for military-action events against the water sector is materially larger in winter than at other times of the year. Frozen ground and infrastructure limit access for emergency repairs. Reduced staffing during holiday periods extends response times. Demand for heating fuel and grid power peaks simultaneously, adding stress to water infrastructure during winter weather. An adversary planning a winter attack on water infrastructure is not executing the same attack as in summer — the consequence envelope, the recovery timeline, and the life-safety risk are all larger. Seasonal amplification is an explicit design parameter in the DBA-MA framework. Security posture, chemical inventory, and emergency response pre-positioning should be adjusted seasonally to reflect the larger consequence envelope during winter.

9. Protective Action and Recovery Architecture

9.1 Threat Detection and Indicator Framework

Elevated threat indicators for the water sector include: Intelligence Community or law enforcement reporting of adversary interest in water sector infrastructure or specific utility SCADA architectures; anomalous ICS network activity including unusual remote-access sessions, unexpected setpoint queries, or lateral movement between enterprise and OT network segments; physical surveillance of water treatment facilities or major pumping stations; unusual aerial UAS activity over water infrastructure; and anomalous chemical procurement patterns that may indicate advance knowledge of supply disruption.

Imminent threat indicators include: confirmed unauthorized remote access to SCADA or HMI systems; simultaneous anomalous parameter changes at multiple points in the treatment or distribution system; confirmed physical intrusion at water treatment facility perimeters; loss of communication with remote terminal units at multiple locations simultaneously; chemical analysis results indicating distribution system contamination; and confirmed kinetic strike on any water sector facility.

9.2 Decision Authority Framework

Water utility operators retain the authority to initiate emergency process shutdowns, chemical dosing adjustments, and distribution system isolations without external authorization. Notification to the state drinking water primacy agency and the EPA is required for events that meet AWIA reporting thresholds. CISA notification is required for confirmed or suspected cyber incidents affecting operational technology. FBI notification is required for contamination events or physical attacks. WaterISAC activation is required for sector-wide information sharing. Coordination of ESF-3 (Public Works and Engineering) is required for infrastructure restoration events. Coordination of ESF-6 (Mass Care, Emergency Assistance, Temporary Housing, and Human Services) is required for extended loss of potable water service at population scale. Coordination of ESF-8 (Public Health and Medical Services) is required for events affecting life-safety facilities — hospitals and dialysis centers — or involving confirmed or suspected distribution of contaminated water. Coordination of ESF-10 (Oil and Hazardous Materials) is required for confirmed contamination events. The ESF activation thresholds for each function correspond to the critical function exceedance conditions defined in Section 4.3.

9.3 Coordination Thresholds

The term “Bright Line” is reserved to FD-BL and is not used in this section: the criteria below are coordination and escalation thresholds, unrelated to the placement of the Bright Line. The following events constitute coordination thresholds that trigger mandatory coordination and, where authority exists, federal response action: confirmed coordinated attack on two or more water system facilities whether within a single service area or regional cluster or across a common device population, a common vendor control platform, or a common integrator or managed-service provider; confirmed distribution of contaminated water to any portion of the served population; confirmed intrusion into treatment facility SCADA or HMI systems with parameter modification capability demonstrated; loss of disinfection residual across a distribution zone serving more than 10,000 population; and loss of water service to any hospital, dialysis center, or equivalent critical dependent facility for any duration.

9.4 Recovery Sequencing Under Military-Action Conditions

Phase 1 — Immediate stabilization (0–24 hours). Manual process control activation for all treatment functions that can be sustained without networked control systems. Chemical dosing verification and adjustment through local manual controls. Distribution system pressure monitoring and isolation of compromised zones. Critical dependent facility notification and alternative supply activation. WaterISAC and CISA notification. Boil-water advisory issuance if disinfection residual integrity cannot be confirmed.

Phase 2 — Short-term bridging (24–72 hours). Emergency generation activation and fuel resupply securing. Chemical inventory assessment and emergency procurement. SCADA forensic investigation initiation — no automated control functions restored until forensic investigation is complete and system integrity is confirmed. System-integrity confirmation for a controller includes comparison of its resident program against an offline gold-copy at the module level, not inspection of the resident program alone. Mutual aid activation from neighboring utilities. Distribution system sampling program initiation to verify water quality.

Phase 3 — Medium-term restoration (72 hours to recovery). Physical damage assessment and repair. SCADA forensic investigation completion and system reconstitution under independent verification. Staged restoration of automated control functions following verification. Return to full treatment capacity — regulatory inspection and compliance verification before lifting any boil-water advisory.

The local sovereignty principle applies throughout all phases. Chemical dosing systems must fail to a conservative safe condition upon loss of networked control. Emergency shutdown systems must actuate passively on process parameter exceedance. Distribution isolation valves must be operable manually at the valve location without SCADA communication. Phase 1 protective actions must be executable by a minimal on-site crew without external communications support. Manual process control presumes authentic indication; where indication integrity cannot be confirmed, operators revert to direct local measurement rather than to displayed values.

10. Facility Class Register and Inheritance Model

10.1 The Inheritance Model

Facility-class documents in the DBA-W series inherit the sector framework's threat taxonomy, bounding methodology, consequence analysis structure, adequacy standards, and command-broker / Bright Line architecture established in Section 5.5. They do not repeat this material. Each facility-class document adds what is specific to its asset profile: facility-class-specific bounding events, consequence chains, design parameters, and recovery architecture, which the sector framework establishes in general terms but does not develop for each asset class.

Each facility-class instrument must include a section explicitly identifying which elements are inherited from this document unchanged, which elements are inherited with facility-class-specific adaptation, and which elements are entirely locally defined. This inheritance map is a

required component of every facility-class instrument at the time of first publication and must be updated at each revision to reflect any changes to this sector framework.

Use cases issued under the sector framework apply the DBA-MA methodology to a specific attack vector or consequence chain within a facility class's scope. Policy companions outline the regulatory, legislative, or institutional actions required to address identified gaps. This modular architecture allows the series to grow as new facility classes are analyzed and new use cases are developed, without requiring revision of the foundational framework documents.

10.2 Facility Class Placement Rationale

The water sector is organized around the physical pathway from source water through treatment and distribution to the end-use tap, and -- for an increasing share of the global and domestic population -- from seawater through desalination to that same distribution system. Facility classes in the DBA-W series are defined by their position in that pathway and by the consequence chain their loss initiates. Water treatment plants (FC1) are placed first because treatment plant outage is the governing consequence event: loss of disinfection capacity converts a supply interruption into a public health emergency within hours. FC1 scope encompasses chemical storage and feed systems as explicit military targets within the facility class: a precision strike against chlorine or sodium hypochlorite storage simultaneously eliminates disinfection capability and creates a chemical hazard in the surrounding area, producing two independent consequence chains from a single strike. The FC1 instrument must address both consequence chains; the chemical-release chain requires a dedicated event category separate from the loss-of-disinfection event category. Distribution pump stations (FC2) are placed second because pump station outage determines whether treated water reaches the population; it is the delivery mechanism whose failure converts available treated water into inaccessible supply. Storage reservoirs (FC3) address the buffering layer that provides the time margin within which treatment plant or pump station recovery must occur; their incapacitation eliminates that margin. Large-scale seawater reverse osmosis desalination facilities (FC4) are placed fourth because their continuous-load characteristic makes them categorically distinct from the other facility classes: SWRO has no passive operating mode, and loss of electric power converts directly to loss of water production on a timescale set by finished-water storage alone, without the natural-source-water buffer available to surface or groundwater treatment plants. In populations dependent on SWRO for their primary supply, military power loss is population-level water failure. FC4 applicability is a site-specific determination based on finished-water storage capacity relative to the population served, not a fixed production-volume threshold; the FC4 instrument specifies the formula the operator applies. Finished-water storage is a named military target within the FC4 facility class: an adversary who understands that SWRO has no passive mode will target finished-water storage before the SWRO plant itself, because destroying storage collapses the only time buffer available and converts SWRO outage from recoverable to immediately catastrophic. Wastewater treatment facilities (FC5) are placed fifth because they sit outside the

source-to-tap pathway but are governed by the same containment principle that underlies the storage reservoir class: their failure consequence is not supply interruption but environmental release and public health exposure from untreated discharge. FC5 carries a distinct regulatory anchor — the Clean Water Act and NPDES permit system — that differs from the Safe Drinking Water Act framework governing FC1 through FC4.

10.3 Facility Class Register

The following table is the current facility class register for the DBA-W series. Facility classes designated as Anticipated require specialist development by qualified water sector engineers and safety analysts. Their inclusion in the register establishes the planned scope of the series; it does not represent completed analytical work.

Designation	Facility Class	NIPP Sector	Status	Primary Regulatory Anchor
DBA-W-FC1	Water treatment plant -- surface and groundwater	Water and Wastewater Systems	Working draft (v0.6)	EPA SDWA; AWIA; state primacy agency requirements
DBA-W-FC2	Distribution pump station	Water and Wastewater Systems	Working draft (v0.6)	EPA SDWA; AWIA; state primacy agency requirements
DBA-W-FC3	Storage reservoir	Water and Wastewater Systems	Working draft (v0.5)	EPA SDWA; state dam safety and primacy agency requirements
DBA-W-FC4	Large-scale seawater reverse osmosis desalination (continuous load; site-specific applicability determination based on finished-water storage capacity relative to population served; see FC4 instrument)	Water and Wastewater Systems	Working draft (v0.4)	EPA SDWA; state primacy agency; USCG (marine intake); FERC (co-located generation where applicable)
DBA-W-FC5	Wastewater treatment facility (municipal; POTW)	Water and Wastewater Systems	Working draft (v0.5)	EPA CWA; NPDES permit system; state environmental agency requirements

10.4 Development Sequence

FC1 (Water Treatment Plant) is the highest-priority facility-class instrument, as the treatment function is the governing public health consequence in the water sector design basis. Its

development requires engagement with water utility operators and state primacy agencies to validate the event categories and adequacy standards against real facility architectures.

FC2 (Distribution Pump Station) follows, addressing the delivery mechanism whose failure converts available treated water into inaccessible supply.

FC3 (Storage Reservoir) follows FC2; its development intersects dam safety engineering and requires coordination with state dam safety programs and EPA.

FC4 (Large-Scale SWRO Desalination) follows FC3; its continuous-load characteristic and dependence on uninterrupted electric supply make it analytically the most demanding facility class in the series, and its development should incorporate the nuclear and hydrogen fuel cell two-tier resilience architecture developed in the Fuel Cell Position Paper (v0.1), Section 1.2, as the second-tier backup power model.

FC5 (Wastewater Treatment Facility) follows FC4. Although its consequence chain is distinct from the source-to-tap classes — its governing concern is environmental release and public health exposure from loss of containment rather than supply interruption — its development shares analytical elements with FC1, particularly the chemical handling and storage systems that are military targets in both classes. Development of FC5 requires engagement with municipal wastewater operators, state environmental agencies, and EPA's Office of Water, and should coordinate with the Clean Water Act primacy structure that differs materially from the SDWA framework governing FC1 through FC4.

11. Path to Regulatory and Legislative Outreach

A sector-wide Design Basis that exists only as a document has no protective effect. The Design Basis established in this document is intended to be incorporated into the regulatory and legislative frameworks governing water sector infrastructure. This section describes that pathway.

11.1 Regulatory Pathway

The Environmental Protection Agency is the federal regulatory authority with the broadest mandate for water-sector resilience under the Safe Drinking Water Act and the AWIA. The Design Basis established in this document is intended to inform EPA's development of updated resilience assessment guidance — potentially as a successor to or supplement to the current AWIA risk and resilience assessment requirements.

The regulatory pathway does not require EPA to mandate the Design Basis in its entirety. A phased approach — beginning with the Design Basis as a voluntary standard, moving to a requirement that AWIA risk and resilience assessments demonstrate consistency with the Design Basis, and ultimately to enforceable performance standards for critical functions — is more achievable and more consistent with the sector's and its regulator's capacity to regulate.

State primacy creates a parallel regulatory pathway. States with mature water sector regulatory programs can adopt the Design Basis as a state-level standard ahead of federal action. Several states have demonstrated a willingness to exceed federal minimum requirements in water sector security. The Design Basis provides those states with a technically grounded standard to reference.

11.2 Legislative Pathway

The AWIA reauthorization cycle provides the principal legislative opportunity. AWIA's current risk and resilience assessment requirements are a process mandate without a performance standard. The legislative pathway would amend AWIA to incorporate a performance standard grounded in the Design Basis — requiring that water systems not merely conduct assessments but demonstrate that their infrastructure meets defined resilience thresholds for Design Basis event categories.

The legislative pathway is more demanding than the regulatory pathway but produces more durable requirements. Process mandates can be satisfied with documentation; performance standards require demonstrated capability. For a sector whose failures produce population-scale public health consequences, the difference matters.

11.3 The Role of the Tiered Framework

The tiered framework — L2 through L4 — transforms the Design Basis from a standard into an implementable program. A regulator or legislator presented with the Design Basis alone has a statement of the conditions that water systems must withstand. Presented with the full tiered framework, they have the standard, the scenario-level analytical basis for each event category, the operational escalation framework for managing conditions in real time, and the monitoring and data-sharing infrastructure requirements. The full framework is the product of regulatory and legislative outreach.

Outreach should not proceed until the tiered framework is sufficiently developed to answer the implementation questions that regulators and legislators will ask. This document establishes the Design Basis and its Military Action extension. The path to outreach runs through the completion of L2, L3, and L4.

References

Federal Statutes

Safe Drinking Water Act (SDWA), 42 U.S.C. §§ 300f et seq.

Clean Water Act (CWA), 33 U.S.C. §§ 1251 et seq.

America's Water Infrastructure Act of 2018 (AWIA), Pub. L. 115-270.

Federal Regulatory Guidance and Directives

Presidential Policy Directive 21 (PPD-21): Critical Infrastructure Security and Resilience. The White House, February 2013.

Environmental Protection Agency. America's Water Infrastructure Act: Risk and Resilience Assessments and Emergency Response Plans — 2023 Operational Guide. EPA 817-B-23-001. Washington, D.C.: EPA, 2023.

Government Plans and Frameworks

Department of Homeland Security and Environmental Protection Agency. Water and Wastewater Systems Sector Specific Plan. Washington, D.C.: DHS/EPA (current edition).

Department of Homeland Security. National Infrastructure Protection Plan: Partnering for Critical Infrastructure Security and Resilience. Washington, D.C.: DHS, 2013.

Standards

American Water Works Association. AWWA J100-21: Risk and Resilience Management of Water and Wastewater Systems. Denver: AWWA, 2021.

International Society of Automation / International Electrotechnical Commission. ISA/IEC 62443 Series: Security for Industrial Automation and Control Systems. Research Triangle Park: ISA (current editions).

Industry and Technical References

Roxey, T.E. "Generative AI Risk in Industrial Control Systems: A Study for Critical Infrastructure Operators, Regulators, and Suppliers." Version 29. Eclectic Technologies, 2026. [Primary source for the criticality bands and the Bright Line concept carried by the CB-Framework. This edition supersedes the prior ISAGCA-attributed "A Study of AI Risks in Critical Infrastructure — Vendor Agnostic," Version 28.]

Cybersecurity Advisories

Federal Bureau of Investigation, Cybersecurity and Infrastructure Security Agency, and Environmental Protection Agency. "Compromise of U.S. Water Treatment Facility." Joint Cybersecurity Advisory AA21-042A. February 2021.

DBA-MA Parent Series

Roxey, Tim. Design Basis Accident — Military Action: International Framework (L1)—version 1.7. Eclectic Technologies, April 2026. [Parent framework establishing the DBA-MA family methodology from which the DBA-W series derives.]

Command Broker Framework Series

Roxey, Tim. Command Broker Framework, Cross-Sector (CB-Framework_Cross_Sector). Eclectic Technologies, 2026. Cited by designation; version state in CB-REG, the Command Broker Family Register. [Canonical cross-sector instrument establishing the broker concept, the Bright Line principle, and the three-tier criticality framework. Section 5.5 of this document is the water sector implementation of that framework.]

Roxey, Tim. Command Broker / Influence Broker Qualification and Onboarding Standard (CB-IB_Qualification_Standard). Eclectic Technologies, 2026. Cited by designation; version state in CB-REG. [Cross-sector qualification methodology and onboarding obligations for CB and IB roles. Governing instrument for command broker qualification in water sector highest-criticality applications.]

INL CCE Methodology

Bochman, Andy, and Sarah Freeman. Countering Cyber Sabotage: Introducing Consequence-driven, Cyber-informed Engineering (CCE). CRC Press, 2021. [Primary reference for the CCE methodology developed at Idaho National Laboratory. The consequence-driven inclusion principle in Section 4.1 of this framework is methodologically aligned with CCE's consequence-first analytical orientation. Organizations that have conducted CCE assessments will find their consequence determinations directly applicable to Design Basis event category scoping under the DBA-W framework.]

Idaho National Laboratory. Consequence-driven Cyber-informed Engineering (CCE). INL/EXT-19-55151. Idaho Falls: Idaho National Laboratory, 2019. [Foundational INL technical report establishing the CCE methodology and its application to critical infrastructure protection.]

Reference Event Instruments

Roxey, Tim. Reference Event 1: Coordinated Cyber Attack on Minnesota Community Water Systems (DBA-W-RE-1 v0.7). Eclectic Technologies, 2026. [Reference event instrument characterizing the July 2026 Minnesota campaign against this framework; source of the indication-integrity, common-substrate, and installed-base findings incorporated in this revision.]

Roxey, Tim. Reference Event 2: Remote Manipulation of the Oldsmar Water Treatment Plant (DBA-W-RE-2 v0.2). Eclectic Technologies, 2026. [Retrospective reference event instrument for the 2021 Oldsmar event, adopting an attribution-agnostic posture; source of the constructed-analog and attribution-independence annotations incorporated in this revision.]

Notes

[1] The practical import of this distinction is significant for water sector utilities. An anomaly detection algorithm already running on a SCADA historian, deterministic within a stated input bound and formally testable in its deployed configuration, meets the carve-out and is outside the Command Broker obligation for its in-bound actions (FD-LD §4.3). That is not a grant of ungoverned autonomy, and it is not a finding that its actions are below the Bright Line. Placement is a property of the individual action and is set by consequence under FD-BL §4.2: an anomaly detector whose erroneous output could produce a consequence at or above the facility's severity threshold produces an above-line action, and it is outside the obligation for that action by the carve-out, not by placement. Where such a component takes autonomous, consequential control action, the case lies in the coverage gap FD-LD §4.8 records, and an in-scope component that changes its thresholds or its bound takes an action placed on its own consequence (FD-BL §4.4). The Gen AI assistant a vendor proposes to add to the HMI for operational decision support is within scope. Each change its output would make once accepted is an action placed on its own consequence, and one that could drive a highest-criticality process parameter to a consequence at or above the threshold is above the line and requires a command broker; its output that would make no change of its own is placed below the line and carries the indication-integrity obligation of FD-BL-D2 wherever it can ground an authorization. The distinction determines which deployed systems require architectural redesign and which do not.

Revision History

Version	Date	Author	Description
0.1–0.6	Mar–Apr 2026	Tim Roxey	Initial working draft through working draft completion. Successive revisions established the sector Design Basis scope and structure; decoupled the MA extension from the base Design Basis document; added Section 5.5 (Command-Broker Architecture and Bright Line) with ISAGCA three-tier criticality framework; formalized the ML / Gen AI terminology distinction; added the References section; added Section 1 statement of intent and Section 2.5 on Engineering Practice as a Pathway to Sector-Wide Standards. Grammarly editorial passes applied throughout.
1.0	April 2026	Tim Roxey	Promoted to the sector framework document, carrying both the non-adversarial Design Basis and the Military Action extension. Sections 7 (Military Action Extension), 8 (Cross-Sector Dependencies), and 9 (Protective Action and Recovery Architecture) were added. Sections renumbered. Section 6 title updated and MA deferral paragraph removed. DBA-MA L1 version reference corrected to v1.5. Document renamed DBA-W-Sector_Framework.
1.1	May 2026	Tim Roxey	Section 5.5 revised to reference CB-Framework_Cross_Sector v0.1 as the canonical cross-sector instrument. Section 5.5.1 attribution updated through the CB-Framework chain. Section 5.5.2 extended to reference CB-IB_Qualification_Standard v0.1. Command Broker Framework Series added to References. I have rebuilt the document as a proper Word file. Cover page corrected to current title and version. References and Revision History moved to back matter.

1.2	May 2026	Tim Roxey	CCE/INL connection added to Section 4.1: new paragraph noting methodological alignment with the Consequence-driven Cyber-informed Engineering (CCE) methodology developed at Idaho National Laboratory. INL CCE Methodology subsection added to References, carrying Bochman and Freeman (2021) and INL/EXT-19-55151 (2019). ESF mapping added to Section 4.3: two paragraphs connecting the four critical functions to their corresponding NRF ESF activation thresholds (ESF-3, ESF-6, ESF-8, ESF-10) and framing Design Basis compliance as a federal response burden reduction instrument. Section 9.2 ESF references expanded from two to four functions with individual activation condition statements and cross-reference to Section 4.3.
1.3	May 2026	Tim Roxey	Section 10 (Facility Class Register and Inheritance Model) added. FC register table established with FC1–FC4 at Anticipated status. Inheritance model and placement rationale paragraphs drafted. Development sequence added. Bright Line Criteria section (9.3) added. Cross-sector dependency mapping (Section 8) completed.
1.4	June 2026	Tim Roxey	Andrew Ohrt (West Yost Associates) reviewer comments incorporated. Section 2.5 (Engineering Practice as a Pathway to Sector-Wide Standards) substantially revised. Section 4 adequacy standards clarified. Grammarly editorial pass applied throughout. Cover page version string corrected.
1.5	June 2026	Tim Roxey	Bright Line spelling corrected throughout ("Brightline" → "Bright Line", two words, title case) in §5.5 heading and body, §5.5.3 and §5.5.4 headings, §6 series document description, §7, §10.1, References annotation, and footnote 1; §9.3 usage untouched (federal coordination triggers, already correct). §10.1 dual-DB architecture declaration added: DBA-W carries a physical DB (this document) and an AI Governance DB (Project 2) simultaneously; §5.5 is the water sector implementation of the AI Governance DB. New §7.1 MA Extension Applicability Determination added at head of Section 7 with three subsections: geolocation (primary factor), strategic value (secondary factor, four indicators), and time-varying property with re-determination triggers; existing §7.1–7.5 renumbered to §7.2–7.6. §10.2 placement rationale revised: FC1 scoping note for chemical storage and feed systems as military targets with dual consequence chain added; FC4 applicability threshold revised from fixed >10 MGD to site-specific storage-to-population formula with finished-water storage named as military target; FC5 (Wastewater Treatment Facility) placement rationale paragraph added. §10.3 register table: FC4 description updated; FC5 row added (DBA-W-FC5, Wastewater treatment facility, Anticipated, CWA/NPDES). §10.4 development sequence: FC5 paragraph added. Revision history v1.3 and v1.4 rows reconstructed.
1.6	August 2026	Tim Roxey	Reference-event disposition revision: incorporates DBA-W-RE-1 (R-1 through R-10) and DBA-W-RE-2 (R-1 through R-5). §1 acknowledges sub-threshold systems and a forthcoming minimum-requirement instrument. §4.1 adds a common-substrate independence test to defense in depth and an attribution-independence statement to consequence-driven inclusion. §4.2.2 broadens the postulated event to controller program/logic modification and falsified indication. §4.2.4 adds the adversarial pressure-loss contamination pathway. §4.3 requires authentic operator visibility. §5.1 names shared remote-support tooling, installed-base and field connectivity, and shared third-party configuration as boundary elements. §5.2 adds the Indication integrity and Protected engineering objects requirements and names remote-support tooling under credential architecture. §5.3 records two reference events, the intact-indication distinction, and the contested Oldsmar attribution. §7.4 annotates the military-action scenario as a constructed analog. §7.5.1 separates capability from direction and records Oldsmar as unattributed. §9.3 extends the coordinated-attack Bright Line beyond geographic contiguity. §9.4 requires offline gold-copy verification and authentic-

			indication fallback. Reference Event Instruments added to References. IAEA L1 parent version reference corrected from v1.5 to v1.7 on the cover and in References.
1.7	August 2026	Tim Roxey	Facility Class Register (§10.3) updated to reflect issued instruments: DBA-W-FC1 (v0.2), DBA-W-FC2 (v0.1), DBA-W-FC3 (v0.1), DBA-W-FC4 (v0.1), and DBA-W-FC5 (v0.1) reclassified from Anticipated to Working draft — all five DBA-W facility classes now issued as working drafts. Register update only; no normative change, and instruments issued under v1.6 remain conformant.
1.8	August 2026	Tim Roxey	Facility Class Register (§10.3) status cells updated to track the West Yost (A. Ohrt) review revisions of three facility-class instruments: DBA-W-FC1 to v0.3, DBA-W-FC2 to v0.2, and DBA-W-FC5 to v0.2 (FC3 and FC4 unchanged at v0.1). Register update only; no normative change, and instruments remain conformant to the framework under which they were issued.
1.9	August 2026	Tim Roxey	Facility Class Register (§10.3) status cells updated to track the Ten States Standards Section 9 closures of two facility-class instruments: DBA-W-FC2 to v0.3 (distribution-pressure floor and booster intrusion floor closed to the Recommended Standards for Water Works, 2022) and DBA-W-FC3 to v0.2 (finished-water storage-volume and turnover parameters closed to the same standard). Register update only; no normative change, and instruments remain conformant to the framework under which they were issued.
1.10	August 2026	Tim Roxey	FD-BL Pass 5 conformance. §5.5 separates the Gate (architecture) from the Bright Line (placement) and states the water-sector conservative implementation as an application of FD-BL that does not vary it; §5.5.3 restores determinism to scope per FD-LD §2.1; §9.3 retitled Coordination Thresholds, reserving “Bright Line” to FD-BL; governing-instruments citation added at §5.5.
1.11	September 2026	Tim Roxey	FD-BL Pass 5 residuals closed. The §5.5.1 explanatory note in Notes asserted that a deterministic anomaly detector “sits below the Bright Line”; the assertion is struck and replaced by the scope carve-out at FD-BL §4.4, with placement left to consequence per FD-BL §4.2 — the carve-out is not the line. The three tier statements in §5.5.1 are conformed to the same reading: determinism decides whether a component is subject to the obligation, not where the line falls for the actions it produces. A stray superseded footer line reading “Version 1.9” was removed from the Revision History page. No change to the design envelope, the facility class register, or any requirement.

1.12	September 2026	Tim Roxey	FD citation form conformed. The governing-instruments paragraph at §5.5 cited all six Foundational Definitions with version tokens; two of those pins had gone stale on the same day (FD-BL-D1 re-issued at v0.4, FD-BL-D2 at v0.3), which is the version-pinned-citation-into-superseded-text defect the citation-form determination exists to end. Tokens dropped; the family is cited by designation and resolves to FD-REG, which is authoritative for FD version state. No requirement changed. A CR-VA conclusion-language sweep of this instrument returned no residual placement assertions.
1.13	October 2026	Tim Roxey	Conformed to the October 2026 Edition of the Foundational Definitions on Project 0's work order of 3 October 2026 (findings P9-01 to P9-05). Scope is stated by the determinism carve-out and cited to FD-LD §4.3, not named by technology class, at §5.4, §5.5, §5.5.2 and §5.5.3 (P9-01). The three tier statements of §5.5.1, the second assessment question of §5.5.3 and Note 1 no longer turn the carve-out into a grant of autonomy, and cite it to FD-LD §4.3 rather than to FD-BL §4.4 or FD-LD §2.1 (P9-02, P9-04). The lowest-tier statement no longer keys autonomous Gen AI action to the tier; band assignment is not placement (P9-03; FD-BL §4.3). The highest-tier statement raises and is unchanged. Influence no longer attaches the obligation to output that makes no change of its own at §5.4, §5.5.4 and Note 1: a recommendation and its acceptance are one action placed on consequence, a person's reliance on output is that person's action, and such output carries FD-BL-D2. The independent-verification demand of §5.5.4 is kept, under FD-BL-D2 I3 (P9-05). Found beyond the screen and conformed in the same pass: Note 1 said the carve-out does not reach an above-line action of a carved-out component, which is false (FD-LD §4.3; FD-BL §4.5), and it now names the coverage gap at FD-LD §4.8. §4.1 carries the DBA-MA-DiD §3.4 disposition step and §3.5 output form by reference (OI-11). CB-Framework_Cross_Sector and CB-IB_Qualification_Standard are cited bare against CB-REG at §5.5, §5.5.2 and References (OI-2), and Influencer Broker reads Influence Broker at §5.5 and in the CB-IB title (OI-5). The reference-event pins in References are repaired to current (DBA-W-RE-1 v0.7, DBA-W-RE-2 v0.2). The Facility Class Register (§10.3) reads the facility-class revisions issued in the same pass on 3 October 2026. Held, not missed: the local-sovereignty sentences at §7.3.4 and §9.4 (P9-06) were read and left as written, pending Project 5's answer on

			whether the DBA-MA construct they name is FD-LS. Revision-history rows were excluded from substitution.
1.14	October 2026	Tim Roxey	Three author decisions of 3 October 2026, taken the same day as v1.13. §5.5 and References follow the CB-Framework's attribution: its criticality bands and the Bright Line concept come from Generative AI Risk in Industrial Control Systems (Version 29), which supersedes the ISAGCA-attributed study (Version 28); the ISAGCA reference is replaced by the Version 29 citation. §4.1 states the author's ruling that trains are credited as independent protection only when robust, diverse and independent, sharing no location, power supply, control system, codebase, vendor or update path; the ruling answers the question DBA-MA-DiD §7.7 holds open and is put to Project 6 for that instrument. §8.1 states the reverse-direction cooling-water obligation of DBA-EN-SF §5.4 and records that FC1 through FC5 carry it, closing OI-12; who bears its cost is open. The Facility Class Register (§10.3) reads the facility-class revisions issued with this one. Revision-history rows were excluded from substitution.
1.15	October 2026	Tim Roxey	Author decision of 6 October 2026, taken with DBA-ESS-RF-FC2 (Response Forces and Stations). §8.3 states that a water facility which is the only firefighting supply for a fire district without an alternate supply carries a tightened criterion for the pressure and flow it delivers to that district, stated jointly with the Emergency Services series. The criterion is not quantified, and its cost is registered under Project 9 OI-17. Revision-history entries are left as records.
1.16	November 2026	Tim Roxey	Cover date corrected. The cover read May 2026, although every version from v1.4 to v1.15 was made between June and October 2026; it now carries the date of this version. Made on the author's instruction of 1 November 2026, before the framework is frozen in the first edition of Project 0's public site of CIKR design basis examples. No other text is changed.